

DK LEARNING

GRADE 10

O/L Computing & Web Foundations

Level: **O/L** • Purpose: Learn before you practise

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In this guide

1. Algorithms and Trace Tables
2. HTML Structure
3. CSS Styling
4. Forms and Links
5. Data and Simple Databases
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 10 activity on the website.

Learning goals

- Trace algorithms using trace tables
- Create simple HTML pages
- Style pages with CSS
- Understand structured data and simple databases

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: In exams, show your trace table. Marks are often given for the working, not only the final answer.

1 Algorithms and Trace Tables

At O/L level, algorithms should be precise enough to implement. Trace variables and conditions carefully and test different inputs. A trace table records each variable after every step.

Pseudocode and flowcharts must use clear, consistent steps.

Key points

- Be precise
- Use a trace table
- Test several inputs
- Include start and end

Worked example

```
n = 1
total = 0
while n <= 3:
    total = total + n
    n = n + 1
# trace: n=1,total=1 | n=2,total=3 | n=3,total=6
```

Your turn

Trace the loop above with $n \leq 4$. What is total?

Check yourself

Explain each term in your own words:

Trace table

Flowchart

My notes

2 HTML Structure

HTML describes the structure of a webpage. A page has `<html>`, `<head>` and `<body>`. Common elements are headings (h1-h6), paragraphs (p), links (a), images (img) and lists (ul, ol).

Elements usually have an opening tag and a closing tag.

Key points

- `<head>` holds page information
- `<body>` holds visible content
- `<a href>` makes a link
- `<img src alt>` shows an image

Worked example

```
<!DOCTYPE html>
<html>
<head><title>My Page</title></head>
<body>
  <h1>Hello</h1>
  <p>Welcome to DK.</p>
</body>
</html>
```

Your turn

Create a page with a heading, a paragraph and a link to another page.

Check yourself

Explain each term in your own words:

Tag

Attribute

My notes

3 CSS Styling

CSS controls presentation such as text colour, spacing and layout. A rule has a selector and declarations. CSS can be inline, internal (in <style>) or external (a .css file).

External CSS is best because one file can style many pages.

Key points

- Selector chooses elements
- Declarations: property: value;
- External CSS reuses style
- Use classes for groups

Worked example

```
h1 { color: #0066cc; font-size: 28px; }  
.card { padding: 16px; border-radius: 12px; }
```

Your turn

Change the page background and the font of paragraphs with CSS.

Check yourself

Explain each term in your own words:

CSS

Selector

My notes

4 Forms and Links

Forms collect information from users using inputs, labels and buttons. Links connect pages. Relative links point to files in your own site; absolute links include a full web address.

Always check links so they do not lead to missing pages.

Key points

- Form uses `<form>` and `<input>`
- Label describes an input
- Relative link: `href="about.html"`
- Absolute link: `https://...`

Worked example

```
<form>  
  <label>Name <input type="text"></label>  
  <button>Send</button>  
</form>
```

Your turn

Add a menu with links to three pages and test each link.

Check yourself

Explain each term in your own words:

Form

Relative link

My notes

5 Data and Simple Databases

Structured data should be stored consistently. Think about valid inputs, meaningful fields and clear relationships between information. A table has fields (columns) and records (rows).

Data should be accurate, up to date and protected.

Key points

- Field = column
- Record = row
- Use consistent formats
- Protect personal data

Worked example

Students table:

ID | Name | Grade

1 | Amal | 10

2 | Nuwani | 10

Your turn

Design a table for a library with four fields.

Check yourself

Explain each term in your own words:

Field

Record

My notes



Practice Questions

Part A - Choose the best answer

1. Which tag holds visible page content?

- A) <body>
- B) <head>
- C) <title>

2. CSS is used for...

- A) styling
- B) storing records
- C) networking

3. A trace table shows...

- A) variable values step by step
- B) a flowchart only
- C) HTML tags

4. creates a...

- A) link
- B) table
- C) image

5. A row in a table is a...

- A) record
- B) field
- C) key

6. Best place to reuse styles?

- A) External CSS
- B) Inline only
- C) Image

Part B - Short answers

1. What is a trace table?

2. What does do?

3. Difference between field and record?

4. Why use external CSS?



Answer Page

Part A

1-A 2-A 3-A 4-A 5-A 6-A

Part B (sample answers)

1. A table recording variable values after each step.
2. Shows an image.
3. Field is a column; record is a row.
4. One file can style many pages.

Mini glossary

Trace table	A table recording variable values step by step
Flowchart	A diagram of an algorithm
Tag	A code word in angle brackets
Attribute	Extra information inside a tag
CSS	Language for styling web pages
Selector	Part of a rule that chooses what to style
Form	A page area for user input
Relative link	A link using a path inside the same site
Field	One kind of data in a table
Record	All data about one item



Activity Preparation

Complete O/L-style algorithm, HTML/CSS and data questions plus a coding challenge. Then open the Grade 10 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 10 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 10 ACTIVITY**

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